

Cixthense

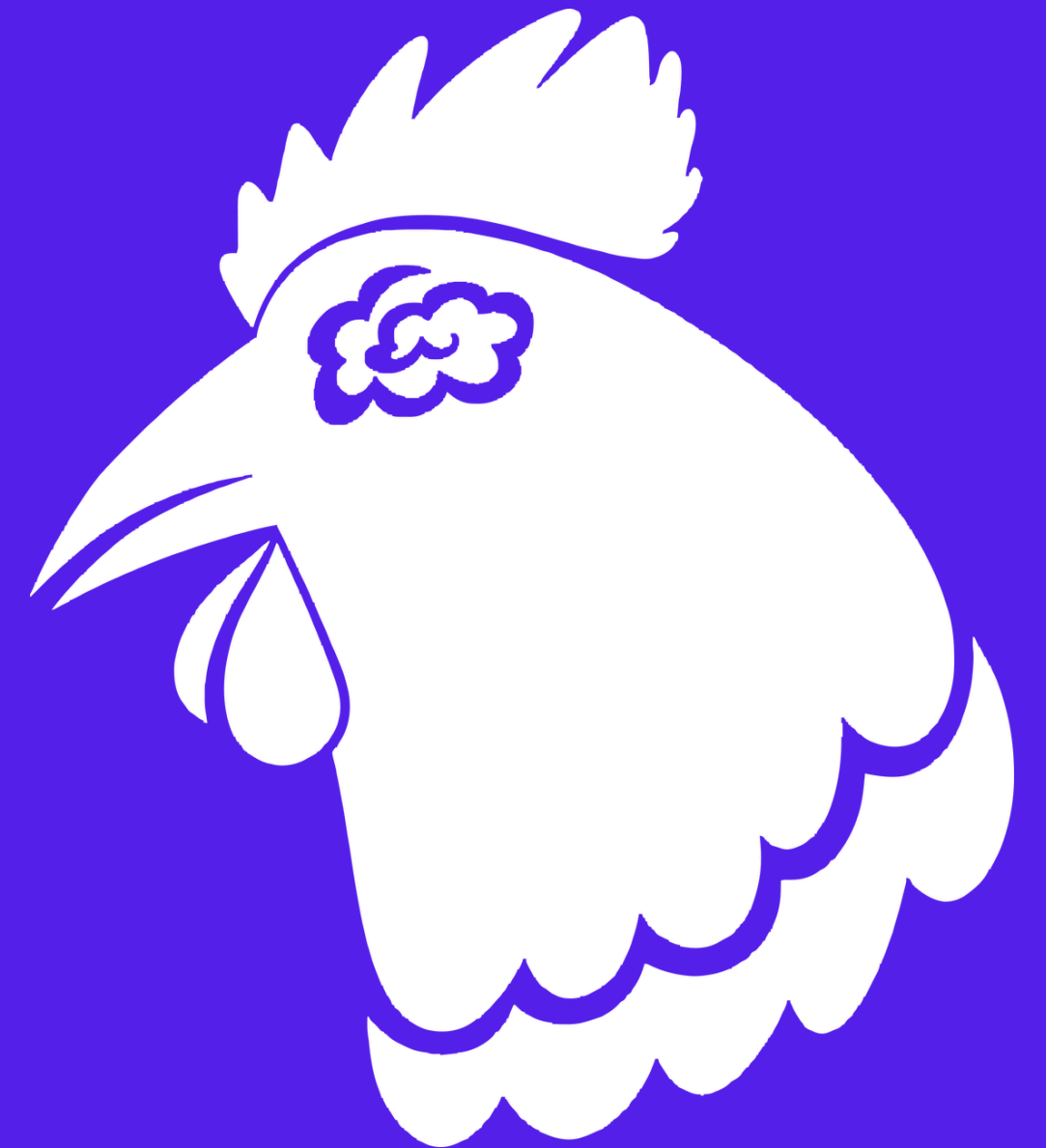


Helping caregivers efficiently sense
and address emotional shifts

Presented by

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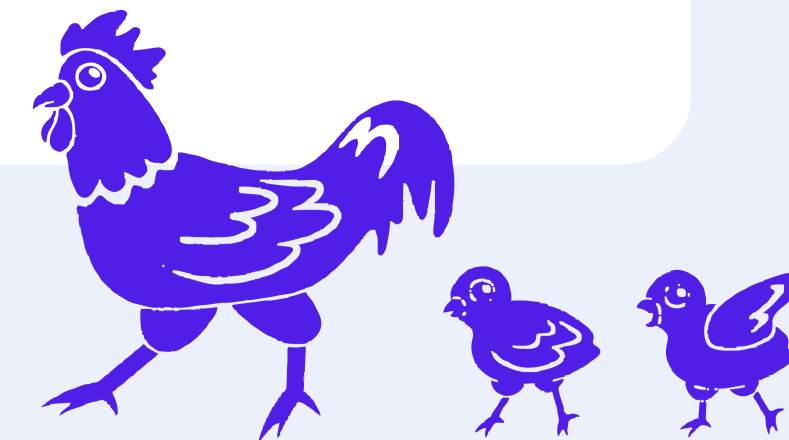


Caregivers are not Mind readers



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How can we help caregivers recognize distress, discomfort, or similar changes in emotional state in **care recipients who have limited communication** to help them **better understand and respond** to the person's needs?



PAIN POINTS

- 60% of caregivers wish to reduce communication breakdowns [1]
- Outward behavior ≠ internal state
- Anger, embarrassment and burnout for both caregiver and recipient [2, 3]
- Safety, medical decisions, maintaining emotional connections [3]

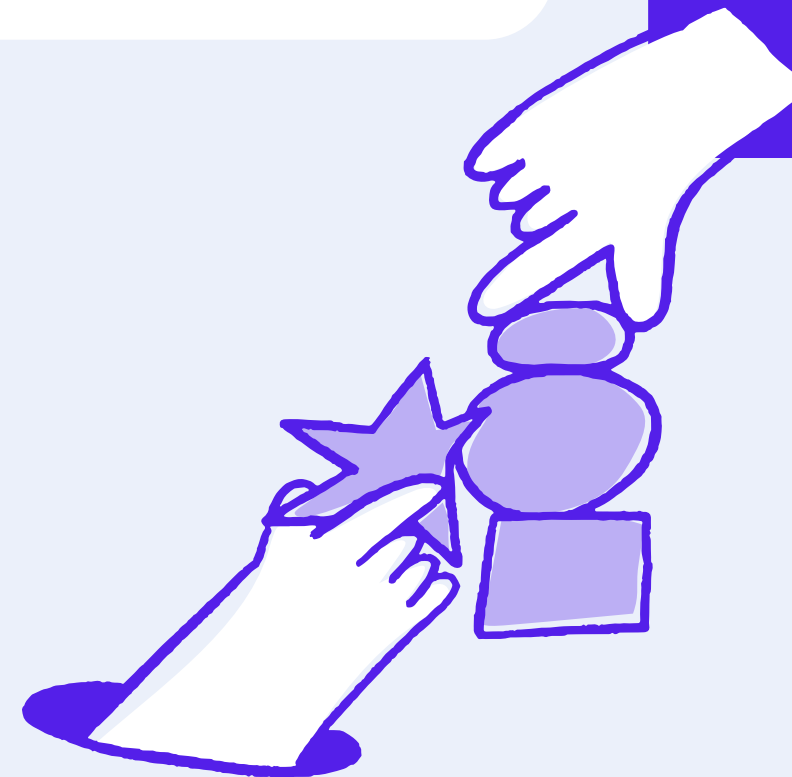


SYSTEM DESIGN

- Calm & intuitive interface
- Additional evidence instead of relying on inference alone
- Recognize distress before it becomes severe

IMPLEMENTATION

- Web-based
- EEG state detection:
Moderate, Elevated, and High Distress
- Pain visualizer
- Previously recorded data to help them respond



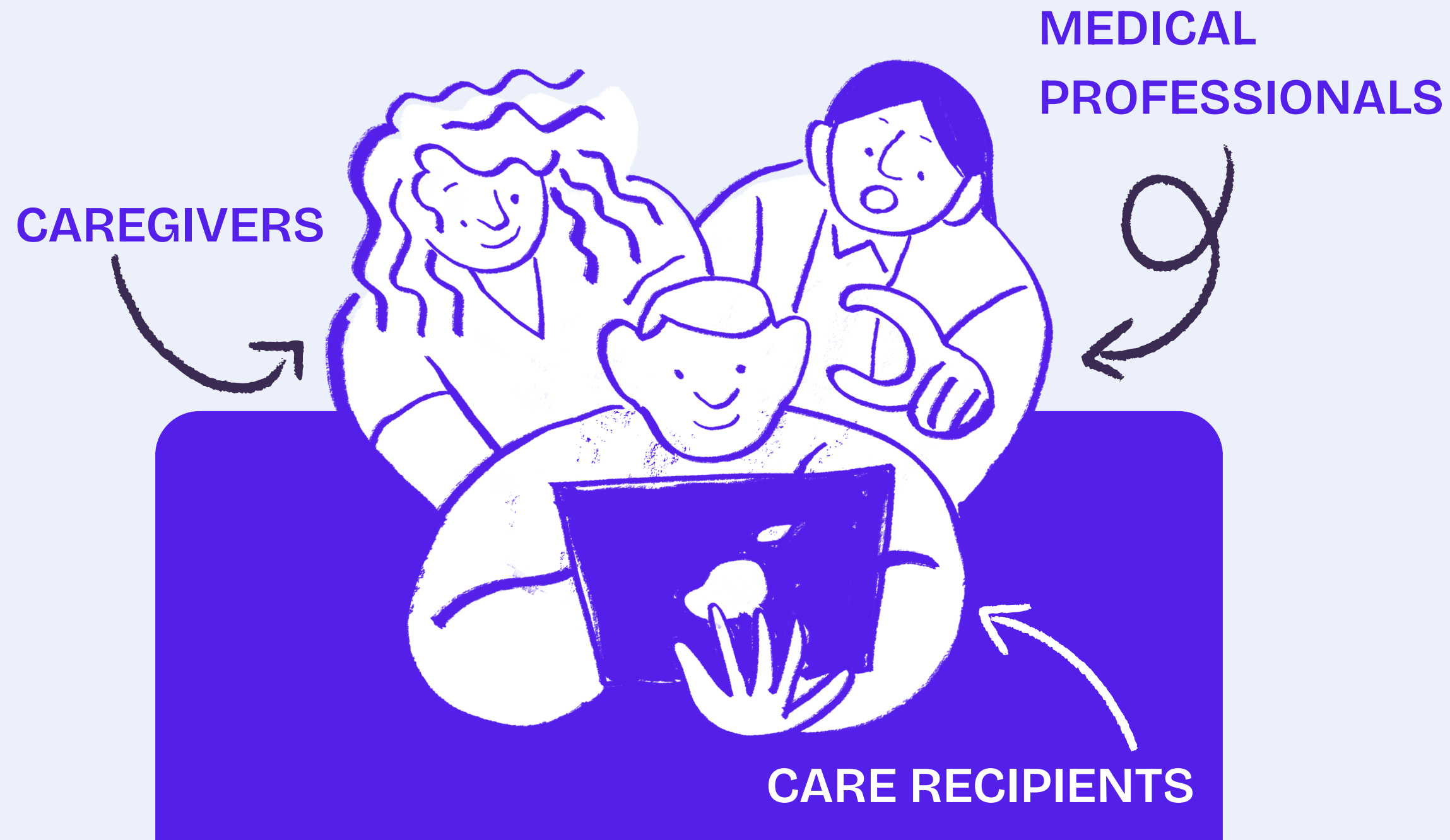
An Objective Approach

A **caregiver-assistance system** that detects notable **changes** in a care recipient's brain activity and alerts the caregiver

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Our Product



See it in Action





KEY INSIGHTS

- **Deviation from baseline** brain activity
- **Valence** (positive/negative) and **arousal** (intensity)
- Similarity to previously recorded periods of distress/overstimulation
- Muscle contractions can interfere

Interpreting the Results

...



We don't have the Full Picture (yet)

STRUGGLES

- Connecting BCI
- Connecting front end to backend
- Can only recognize arousal/valence, not actual emotions
- Implementing design thinking

FUTURE FEATURES

- External sensors + AI insights & resources
- Mobile app
- ML model trained on past data

OTHER APPLICATIONS

- Babies
- Pets/farm animals (chickens)*
- Personal emotional regulation



Thank You!

Questions?



Works Referenced

[1] F. Pierson, "What Are Common Challenges for Caregivers and Care Partners?" Leader Live, May 2025, doi: 10.1044/2025-0507-nslhm-caregiver-poll-aud-slp/full/.

[2] Grit Psychology, "Understanding Caregiver Burnout: The Psychology Behind the Struggle - Grit Psychology," Grit Psychology. Accessed: Sep. 19, 2026. [Online]. Available: <https://gritpsychology.com/understanding-caregiver-burnout-the-psychology-behind-the-struggle/>

[3] Coffee with America, The Emotional Reality of Caregiving and Communication Challenges, (Jun. 05, 2026). Accessed: Sep. 19, 2026. [Video]. Available: <https://www.youtube.com/watch?v=cvMQWIHIBu8>

[4] K. N. Eberle, M. P. Martin, S. Shah, R. D Malheiros, K. A. Livingston, and K. E. Anderson, "A Novel Non-Invasive Method for Evaluating Electroencephalograms on Laying Hens," Poultry Science, vol. 97, no. 3, pp. 860–864, Mar. 2018, doi: 10.3382/ps/pex391.